



Mastering Similar Triangles!

(and how to hack your
geometry exam ✨)



Let's unlock the secrets...

CONGRUENT



Same Shape + Same Size



SIMILAR



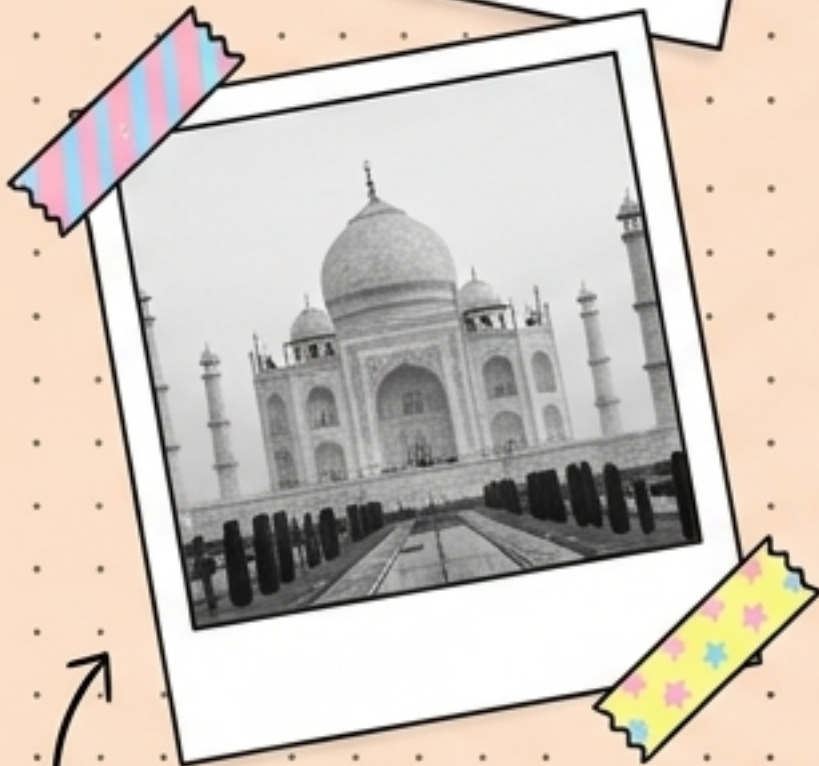
Same Shape + **DIFFERENT** Size



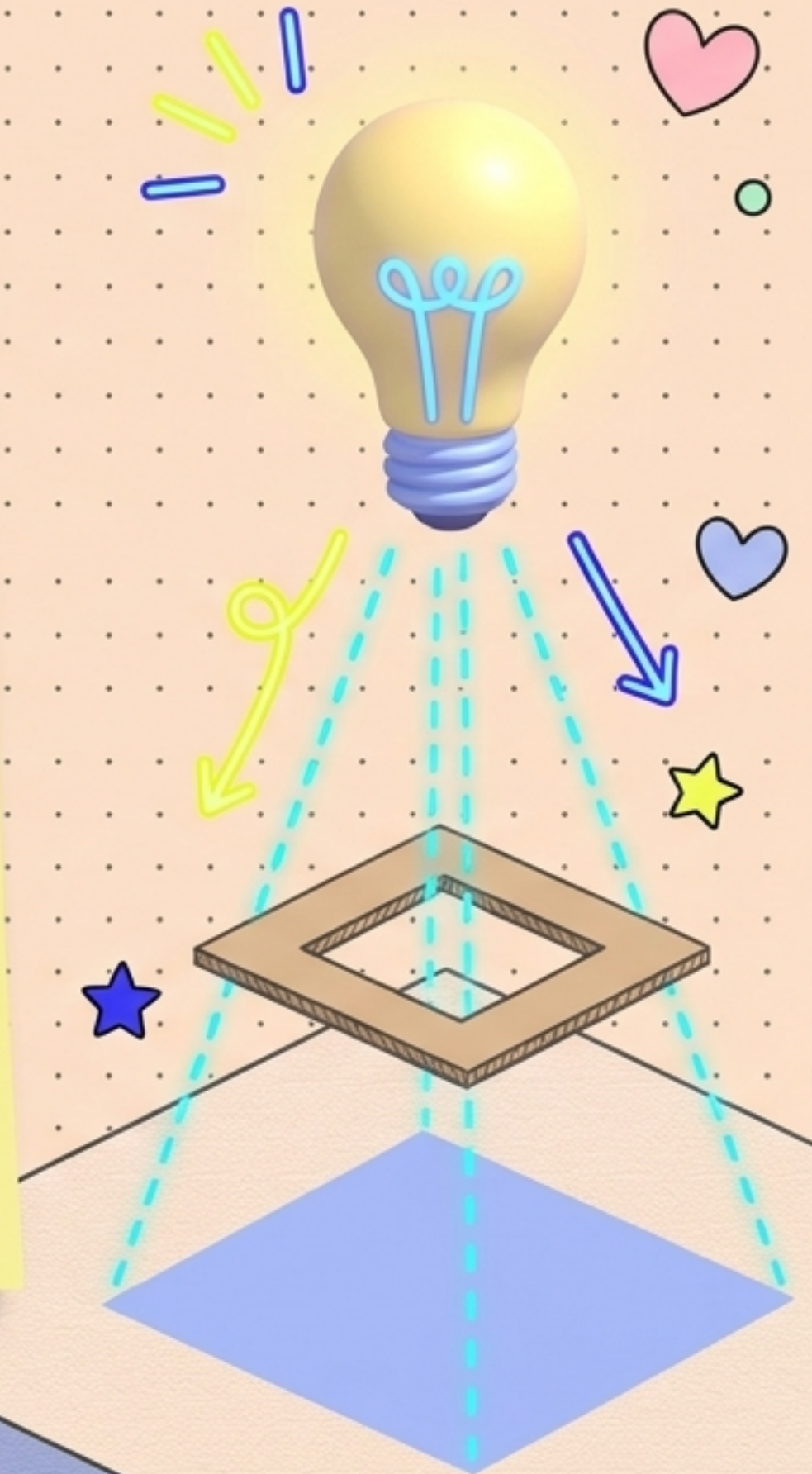


The 2 Golden Rules of Similarity

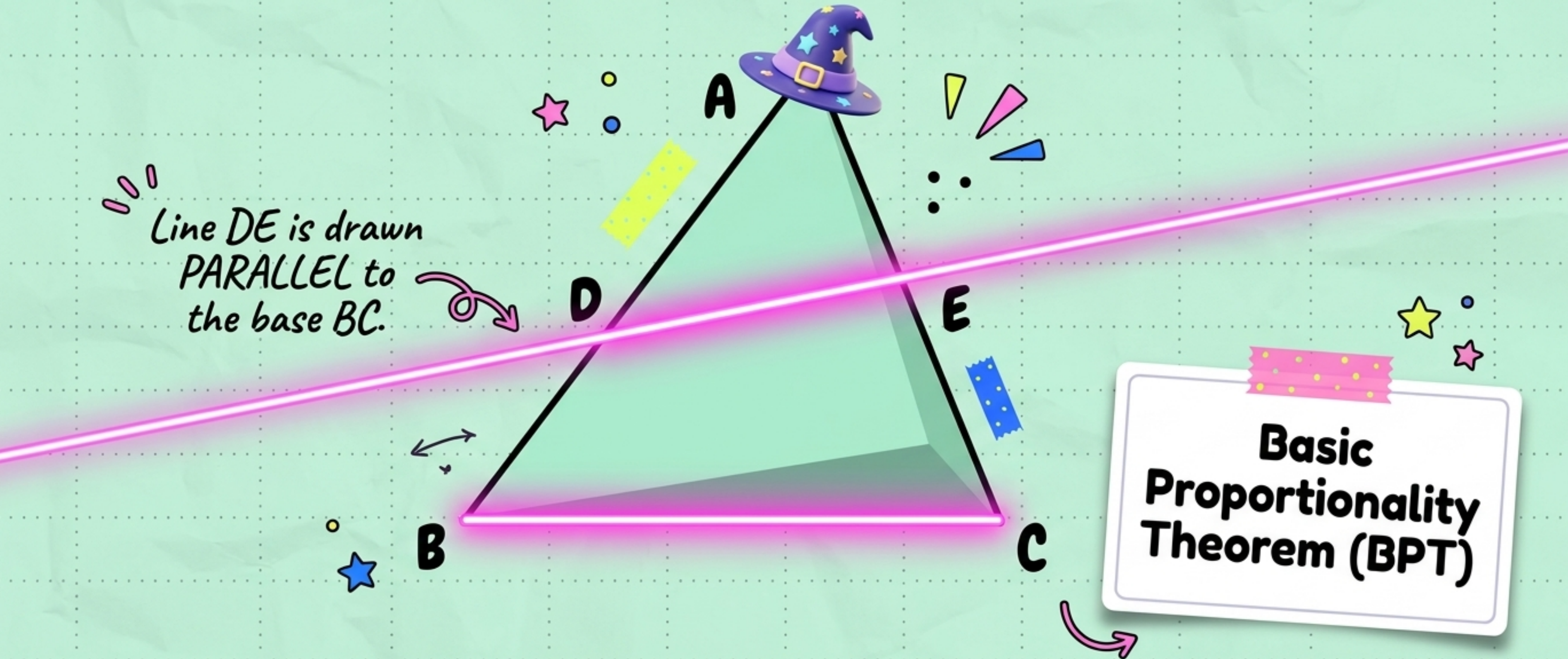
- ✓ Corresponding Angles are EQUAL.
- ✓ Corresponding Sides are PROPORTIONAL (The Scale Factor!)



Real life examples!

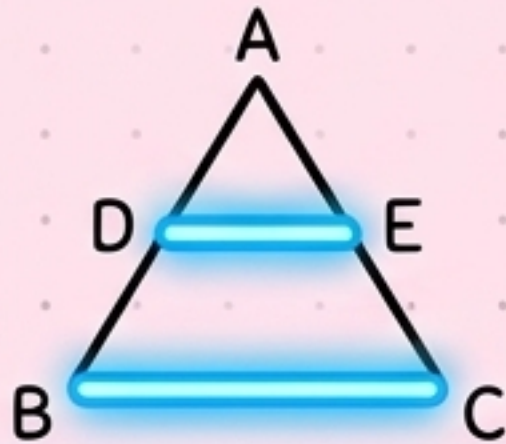


Enter the Triangle!

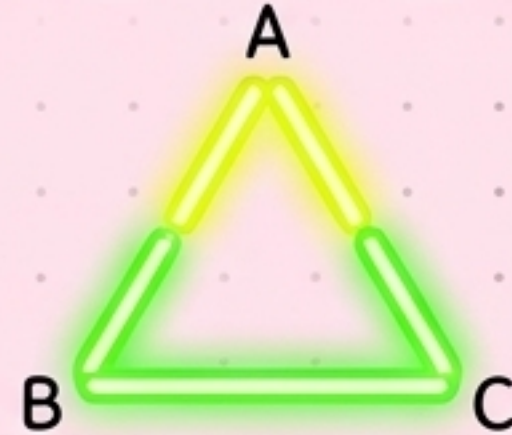


The BPT Flowchart

IF $DE \parallel BC$



$$\frac{AD}{DB} = \frac{AE}{EC}$$



The line divides the sides in the EXACT same ratio!

The Reverse Spell

If the ratios
are identical...

$$\frac{AD}{DB} = \frac{AE}{EC}$$



...Then the
lines **MUST** be
parallel!

$$DE \parallel BC$$

This is Theorem 6.2 (The Converse of BPT).



The Similarity Cheat Sheet

AAA
(Angle-Angle-Angle)

1



SSS
(Side-Side-Side)

2



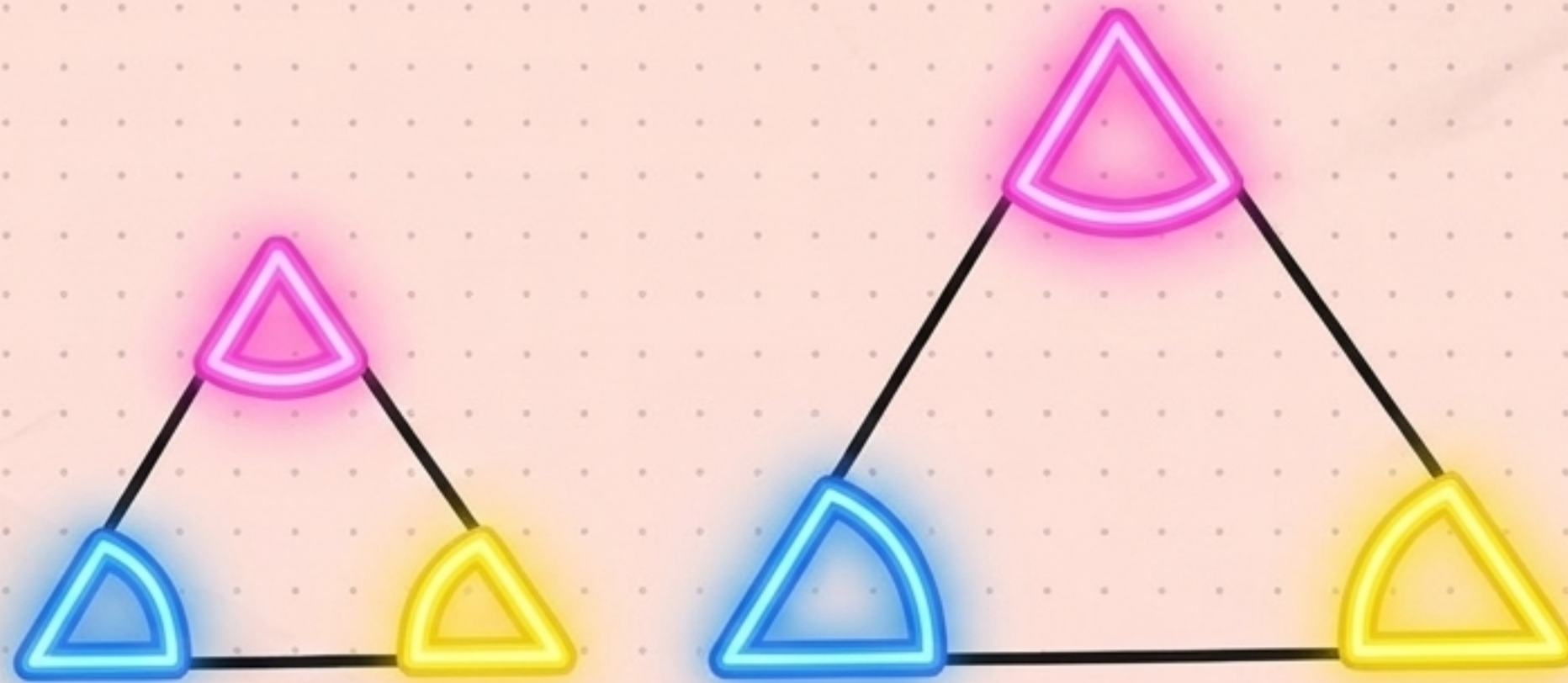
SAS
(Side-Angle-Side)

3



You only need *ONE* of these 3 shortcuts to prove triangles are similar!

Shortcut #1: AAA

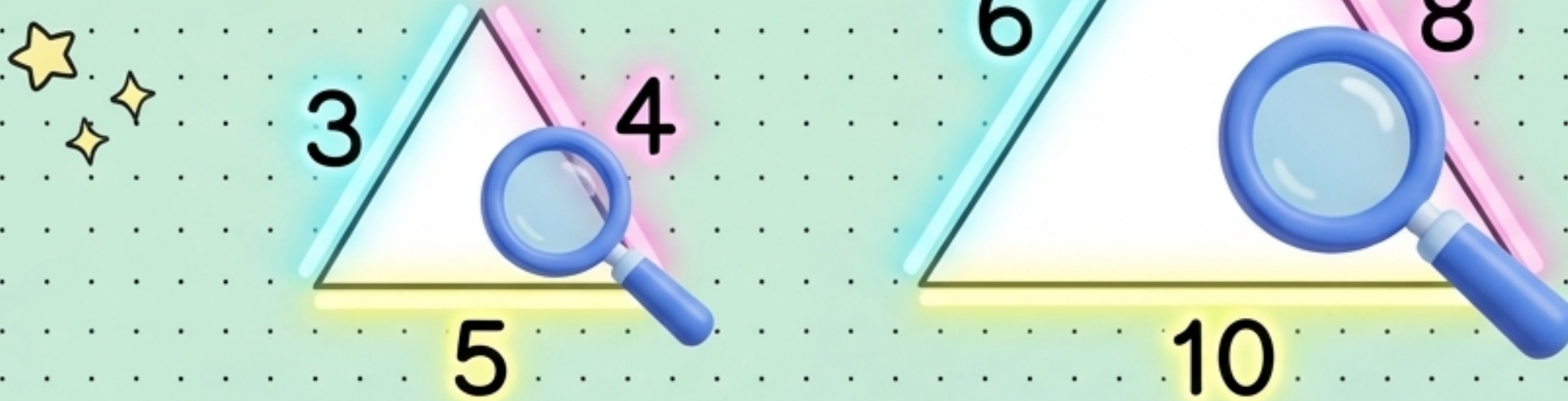


If all corresponding angles match,
the triangles are similar.

✨ HACK: AA is enough! Because
inside of a triangle
always equals 180° ,
if 2 angles match,
the 3rd one has to
match automatically!



Shortcut #2: SSS



$$\frac{3}{6} = \frac{1}{2} \quad | \quad \frac{4}{8} = \frac{1}{2} \quad | \quad \frac{5}{10} = \frac{1}{2}$$

All 3 sides are perfectly proportional
(in the same ratio).

Shortcut #3: SAS



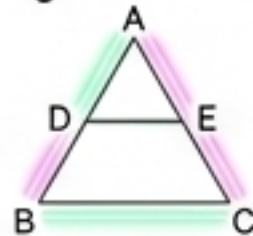
Two proportional sides +
The INCLUDED Angle.

*Warning: The angle MUST be trapped directly
between the two sides you are measuring!*

The Ultimate Exam Cheat Sheet

BPT

$$\frac{AD}{DB} = \frac{AE}{EC}$$



AAA

Angles =
equal



SSS

Sides =
proportional



SAS

Sides
proportional
+
Included angle



You've got this! Save this post for your midterms! 💖